

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Space Age, we delve into the evolution of Mars rovers; and Origins covers the history of light switches and its evolution.



Rapid testing for cancer

A team of international researchers from the UK, US and Germany built a new easy-to-use test that could quickly diagnose non-infectious diseases like heart attacks and cancers. <https://dgit.in/CancerTests>

WHAT'S NEW

Pipeline for NASA

The US space agency has asked university students to design a metal production pipeline on the Moon – from extracting metal from lunar minerals to creating structures and tools. The ability to extract metal and build needed infrastructure on the Moon advances the Artemis Programme goal of a sustained human presence on the lunar surface, NASA said in a statement. The 2023 Breakthrough, Innovative and Game-Changing (BIG) Idea Challenge invites university students to tackle some of the most critical needs facing space exploration and help create the mission capabilities that could make new discoveries possible. Student teams will develop innovative ways to extract and convert metals



from minerals found on the Moon, to enable metal manufacturing. The NASA-funded challenge

provides development awards of up to \$180,000 to up to eight selected teams. The availability of derived metals on the Moon would allow infrastructure needed for a lunar base – including pipes, power cables, landing pads, transport rails, and pressure vessels to contain volatiles like fuel – to be made locally using additive manufacturing, or 3D printing.

<https://dgit.in/NASAPipeline>

NASA discovers CO2 in an exoplanet system for the first time

NASA's James Webb Space Telescope has captured the first-ever clear evidence for carbon dioxide in the atmosphere of a planet outside the solar system. The finding offers hope that in the future, Webb may be able to detect and measure carbon dioxide in the thinner atmos-



pheres of smaller rocky planets. This observation of a gas giant planet named 'WASP-39 b' provides important insights into the composition and formation of the planet. Previous observations from other telescopes, including NASA's Hubble and Spitzer space telescopes,

revealed the presence of water vapour, sodium, and potassium in the planet's atmosphere. "Detecting such a clear signal of carbon dioxide on WASP-39 b bodes well for the detection of atmospheres on smaller, terrestrial-sized planets," said Natalie Batalha of the University of California at Santa Cruz, who leads the team. <https://dgit.in/JWSTAgain>



Using smart-phones can improve memory

A new study has found that smartphones and other digital devices could actually help improve memory skills rather than causing people to become lazy or forgetful. Huh, quite a revelation! <https://dgit.in/PhoneGood>



Mimicking human brains

A team of US researchers have developed a flexible, stretchable computing chip that processes information by mimicking the human brain. The device, described in the journal Matter, aims to change the way health data is processed. <https://dgit.in/HumanAI>



Blenderbot 3 beta test begins

BlenderBot 3 has been released to public users in the US for beta testing. The bot is a prototype based on Meta's previous work with large language models (LLMs) and is trained on massive text datasets to find statistical patterns and produce language. <https://dgit.in/BB3>